24V input on GIGA shield | Local 3.3V display conversion | Molex 15018-0563 Type-A 60-way FFC, 102mm nominal

ribbon

Datei: ribbon.kicad_sch

buffer1

Datei: buffer1.kicad_sch

buffer2

Datei: buffer2.kicad_sch

buffer3

Datei: buffer3.kicad_sch

buffer4

Datei: buffer4.kicad_sch

touch

Datei: touch.kicad_sch

power_input

Datei: power_input.kicad_sch

power12

Datei: power12.kicad_sch

headers

Datei: headers.kicad_sch

Supply Present

power:PWR_FLAG

Supply Present

power:PWR_FLAG

Supply Present

power:PWR_FLAG

ProtoMate

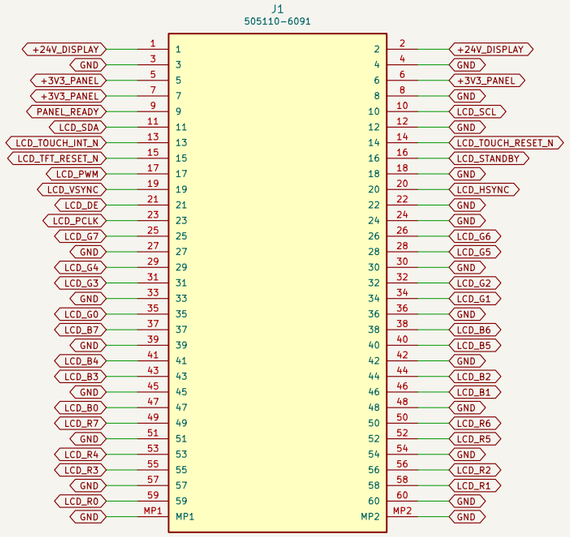
Sheet: /

File: giga-adapter.kicad_sch

Title: giga-adapter / Smartwin display interface

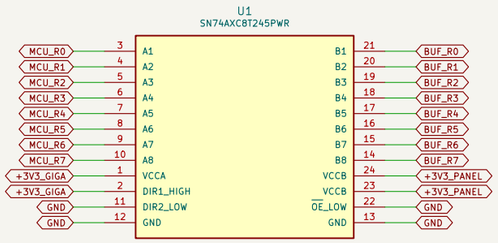
Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 1/10

60-way Molex 15018-0563 Type-A / 505110-6091 sockets; 0.5mm pitch, 0.30mm mating thickness; GIGA n <-> display 61-n.

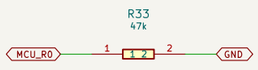


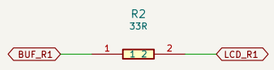
ProtoMate		
Sheet: /ribbon/ File: ribbon.kicad_sch		
Title: giga-adapter / ribbon		
Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 2/10

DIR1 = HIGH; DIR2 = LOW; OE = LOW. 33R series termination; 47k input bias. All channels A -> B.

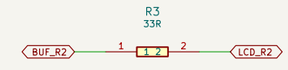




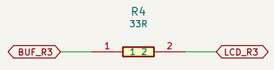




















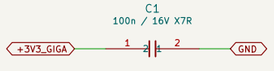














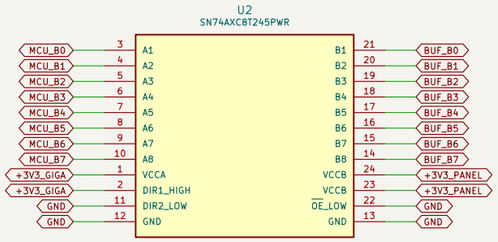
ProtoMate

Sheet: /buffer1/
File: buffer1.kicad_sch

Title: giga-adapter / buffer1

Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 3/10

Same isolation and termination as U1. Both supplies must be valid for transmission.





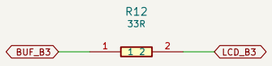




















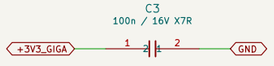


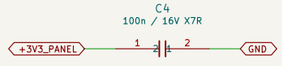












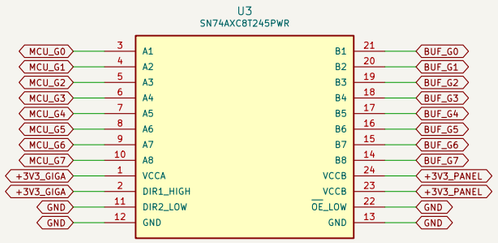
ProtoMate

Sheet: /buffer2/
File: buffer2.kicad_sch

Title: giga-adapter / buffer2

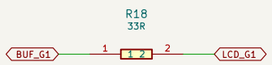
Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 4/10

RGB24 physical interface. All eight green bits are connected.







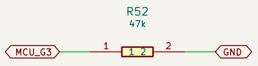


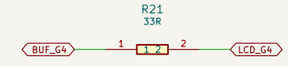






























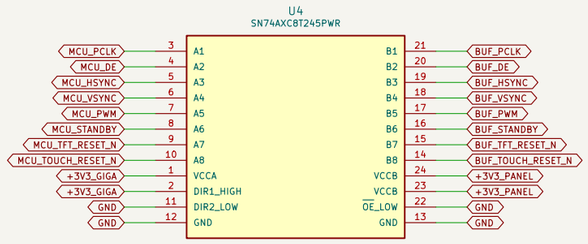
ProtoMate

Sheet: /buffer3/
File: buffer3.kicad_sch

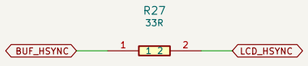
Title: giga-adapter / buffer3

Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 5/10

PCLK, DE, HSYNC, VSYNC, PWM, STANDBY and active-low resets. PCLK buffered by the same family as RGB.







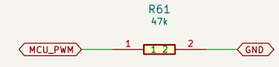










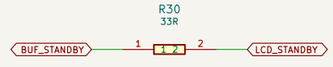








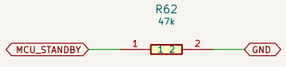














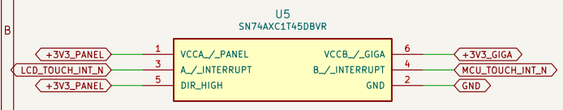
ProtoMate

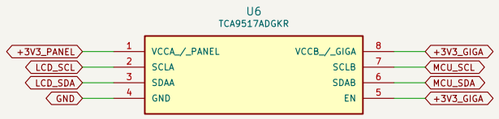
Sheet: /buffer4/
File: buffer4.kicad_sch

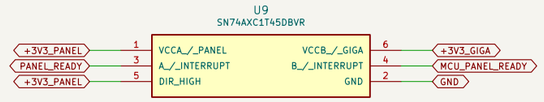
Title: giga-adapter / buffer4

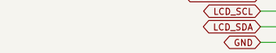
Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 6/10

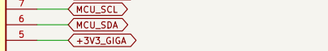
TCA9517 A = panel; B = GIGA. GIGA R26/R27 already provide 5.1k pull-ups. AXC1 direction A -> B.











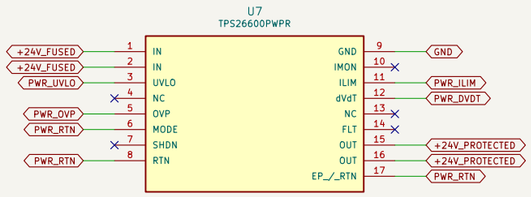
ProtoMate

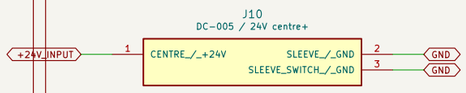
Sheet: /touch/
File: touch.kicad_sch

Title: giga-adapter / touch

Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 7/10

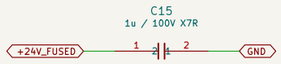
J10: 5.5/2.1mm DC-005, centre +24V; sleeve and switch GND. UVLO ~18.2V; OVP ~28.1V; limit ~1.49A. RTN is NOT GND.









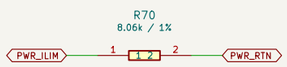


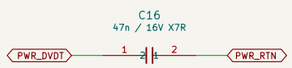


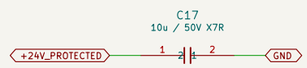














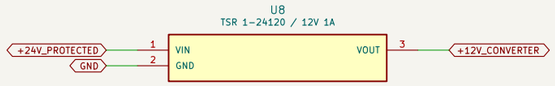
ProtoMate

Sheet: /power_input/
File: power_input.kicad_sch

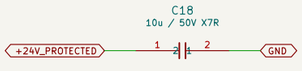
Title: giga-adapter / power_input

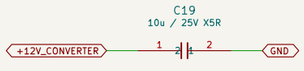
Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 8/10

U8: 24V -> 12V / 1A. D3 blocks USB-only reverse feed into U8. F2 protects the two parallel 24V ribbon conductors.















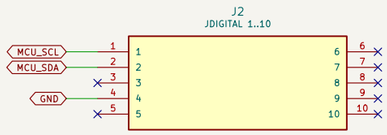
ProtoMate

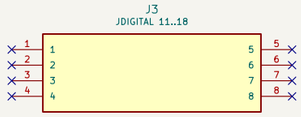
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File: power12.kicad_sch

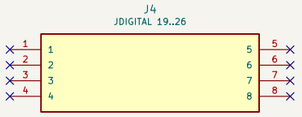
Title: giga-adapter / power12

Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 9/10

NC means no adapter trace. Socket still passes that GIGA contact straight through. Do not populate other header variants.



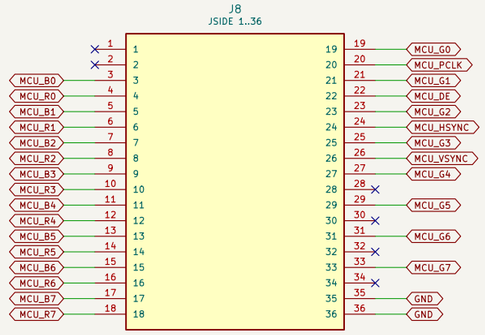


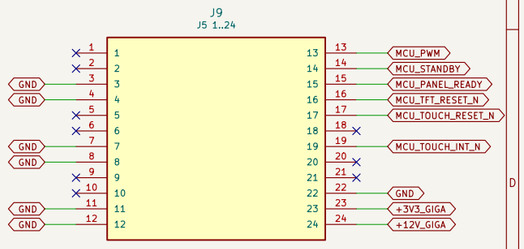












ProtoMate

Sheet: /headers/
File: headers.kicad_sch

Title: giga-adapter / headers

Size: A3	Date: 2026-10-22	**Rev: E**
KiCad E.D.A. 10.0.6		Id: 10/10